Experiment 1: GCD discovery + frozen CLIP grounding

Results on CUB-200-2011 · training seed 0 · report of 10 November 2026 · code commit `811ebfb`

Summary

- **The discovery stage reproduces GCD.** On the unlabelled training images (known K = 200) the v2 accuracy is 50.8 / 55.8 / 48.2 (All / Old / New), against 51.3 / 56.6 / 48.7 in the GCD paper; three clustering seeds give 49.7–50.8 All.
- **CLIP names 40.0% of discovered images correctly** with the complete 200-name vocabulary (95% CI 35.2–45.4; Old 48.9, New 35.6). Because each cluster gets one name, the best achievable with these clusters is 52.2%.
- **Rejection ("unmapped") helps only when many true names are missing.** With the complete vocabulary it costs accuracy; with 20 of 200 names it raises end-to-end accuracy from 7.5% to 84.4%, mostly by rejecting.
- **The simplest score separates present from missing names best** (max similarity, AUROC 0.65 at 100 names), but separation is weak at large vocabularies. When a class's name is swapped for a similar-looking class, 74% of its images take the substitute's name.
- **Clustering errors are image-level.** In the development pool, misclustered images are farther from their GCD centre and less typical of their cluster's similarity pattern; clusters that contain errors look like the others.

1. What was run

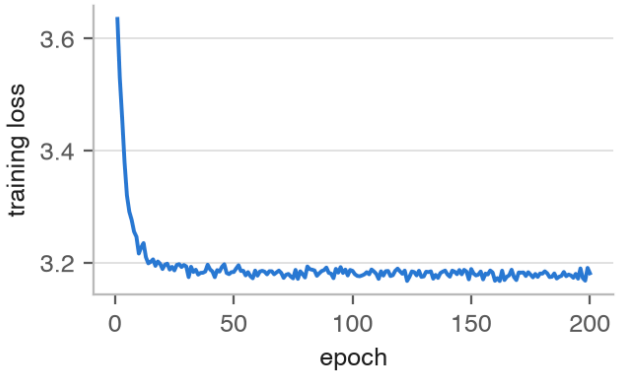
1 · Data	2 · Discovery	3 · Representation	4 · Grounding	5 · Evaluation
CUB-200-2011, SSB split: 100 known, 100 novel classes; half of the known-class training images are labelled.	Upstream GCD: DINO ViT-B/16, contrastive training 200 epochs (final epoch), semi-supervised k-means.	Frozen OpenAI CLIP ViT-B/32 (and B/16) image and text embeddings.	Each cluster gets the name with the highest mean similarity, or is rejected ("unmapped") by a calibrated score.	Image-level, on the unlabelled images; vocabularies of 200, 150, 100, 50 and 20 names.

All settings were fixed before the run (`PROTOCOL.md`). Labels are kept in separate tables by permission: labelled anchors (1498 images) for training; held-out official test images of the known classes for choosing prompts and the softmax temperature (development pool, 1433) and for setting rejection thresholds (calibration pool, 1451); the 4496 unlabelled training images (1499 known-class, 2997 novel-class) are used only for the final evaluation. Evaluation is transductive: discovery saw these images without labels.

Incomplete vocabularies are random draws of 150, 100, 50 or 20 class names (5 draws each), shared by all clusters. An image whose true name is in the vocabulary is correct only if its cluster is accepted with that name; an image whose name is missing is correct only if its cluster is rejected (**end-to-end accuracy**). Thresholds are calibrated once across all vocabulary sizes. Intervals are 95% bootstrap intervals that resample discovered clusters together with their images.

2. Checks before the results

Check	Result
Labelled/unlabelled split equals the upstream GCD code's	passed: 1,498 labelled, 4,496 unlabelled
All images present and decodable	8878 images decoded; 0 missing, 0 corrupt
Training completed	200 epochs in 9.62 h on a Tesla T4, fp32 as upstream
CLIP matches OpenAI's official package (ViT-B/32)	min. cosine 1.000000 (images), 1.000000 (texts); tokens identical
The pilot's CLIP setup (GELU activation)	agrees with the official model on only 66% of zero-shot labels
Software tests	26 tests pass (partitions, k-means repairs, metrics, resumption, label isolation)



Training loss per epoch (seed 0). It drops in the first ~10 epochs and is flat afterwards.

Throughput on a T4 (batch 128 images, 256 views): 40.0 images/s in fp32, 113.5 with mixed precision. Peak GPU memory 4.76 GB. The earlier pilot's CLIP numbers should not be compared with these: its activation function differed from the trained model's.

3. Discovery (GCD) on the unlabelled images

Number of clusters	Clustering seed	All	Old	New	NMI	ARI	Singletons
known K = 200	0	50.8	55.8	48.2	0.783	0.381	39
known K = 200	1	49.7	52.6	48.2	0.782	0.381	44
known K = 200	2	49.9	53.6	48.1	0.782	0.386	42
estimated K = 199	0	49.2	51.1	48.2	0.784	0.383	40
GCD paper (arXiv v2, Table 3)		*51.3*	*56.6*	*48.7*			

v2 accuracy: one Hungarian assignment of clusters to classes over all unlabelled images, with Old/New measured afterwards. K was estimated at 199 from the labelled images only (true 200). The Hungarian assignment is used for these numbers only, never for naming.

Cluster quality limits naming: the median cluster purity is 0.71, and 68 of 200 clusters have no majority class; they hold 2,524 of the 4,496 unlabelled images (56%).

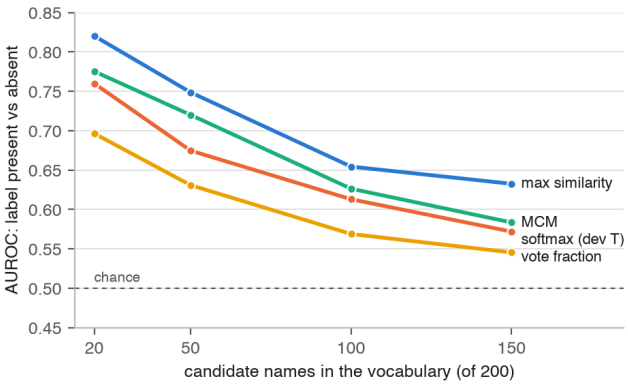
4. Naming clusters with the complete vocabulary

CLIP model and prompts	Image-weighted	Class-macro	Old	New
ViT-B/32, forced top-1 (80 prompts)	40.0	42.4	48.9	35.6
ViT-B/32, MCM prompt	40.2	43.6	52.4	34.1
ViT-B/16, forced top-1 (80 prompts)	40.5	43.1	49.4	36.0
ViT-B/16, MCM prompt	38.9	41.7	48.4	34.2
Ceiling: every cluster named after its majority class	*52.2*			

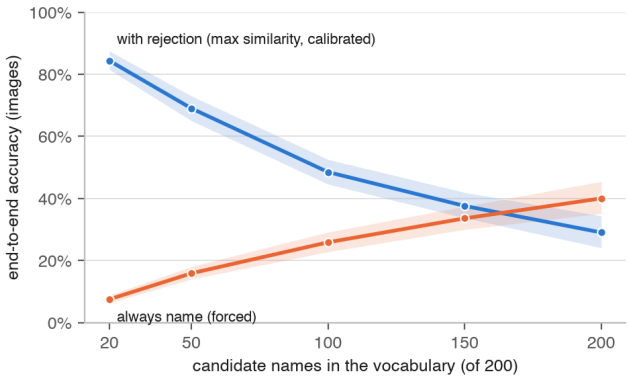
Accuracy (%) of the cluster's name for each of its images. The development pool chose the 80-template prompt ensemble over the single prompt for both models (T* = 0.02).

CLIP reaches about three quarters of what the clusters allow. Known-class images are named better than novel-class ones, partly because known-class clusters are purer. ViT-B/16 is not better than ViT-B/32 here.

5. Naming with rejection when true names are missing (CLIP ViT-B/32)



How well each score separates images whose true name is present from images whose name is missing (0.5 = chance).



End-to-end accuracy with and without rejection; shaded bands are 95% cluster-bootstrap intervals.

Vocabulary	AUROC	Named correctly (name present)	Accepted	Correct among accepted	Missing-name images accepted	End-to-end, with rejection	End-to-end, forced	End-to-end Old / New
200 (complete)	—	40.0	64.7	44.9	—	29.0 [24.1, 34.4]	40.0	34.6 / 26.3
150	0.63	45.2	56.9	40.7	43.6	37.6 [33.8, 41.9]	33.6	43.8 / 34.5
100	0.65	51.9	46.1	36.6	36.9	48.5 [44.6, 52.5]	25.9	53.5 / 46.0
50	0.75	63.9	29.2	32.9	20.8	69.0 [65.1, 73.0]	15.9	73.6 / 66.8
20	0.82	78.9	15.2	29.9	11.4	84.4 [81.6, 87.5]	7.5	86.2 / 83.5

Max-similarity score with its calibrated threshold (τ maximising calibration end-to-end accuracy), mean over 5 vocabulary draws per size. "Named correctly" ignores rejection. Percentages except AUROC.

All rejection scores

Score	AUROC, 150	100	50	20	Difference vs max similarity at 100 (AUROC points)
max similarity	0.63	0.65	0.75	0.82	reference
top-1 – top-2 margin	0.54	0.55	0.59	0.68	-10.1 [-13.5, -6.6]
softmax, pilot T = 0.01	0.56	0.58	0.63	0.72	-7.2 [-10.1, -4.2]
softmax, dev T* = 0.02	0.57	0.61	0.67	0.76	-4.1 [-6.6, -1.8]
MCM (T = 1)	0.58	0.63	0.72	0.78	-2.8 [-5.2, -0.5]
MCM, per-image mean	0.60	0.64	0.73	0.78	-1.2 [-3.9, +1.5]
vote fraction	0.55	0.57	0.63	0.70	-8.5 [-11.3, -5.8]
softmax entropy	0.59	0.63	0.72	0.78	-2.5 [-4.9, -0.2]
vote entropy	0.54	0.58	0.64	0.71	-7.2 [-9.9, -4.4]

Paired differences use the same bootstrap resamples. Softmax-based scores, including MCM, change scale with the vocabulary size; because thresholds are calibrated across all sizes at once, MCM ends up rejecting nearly every cluster (Section 8).

Similar substitutes

10-name vocabulary built around one target class	Images given the target's name	Given the substitute's name	Accepted
true name present	85	—	46
true name replaced by a random class	0	9	6
true name replaced by the most similar class name	0	74	25

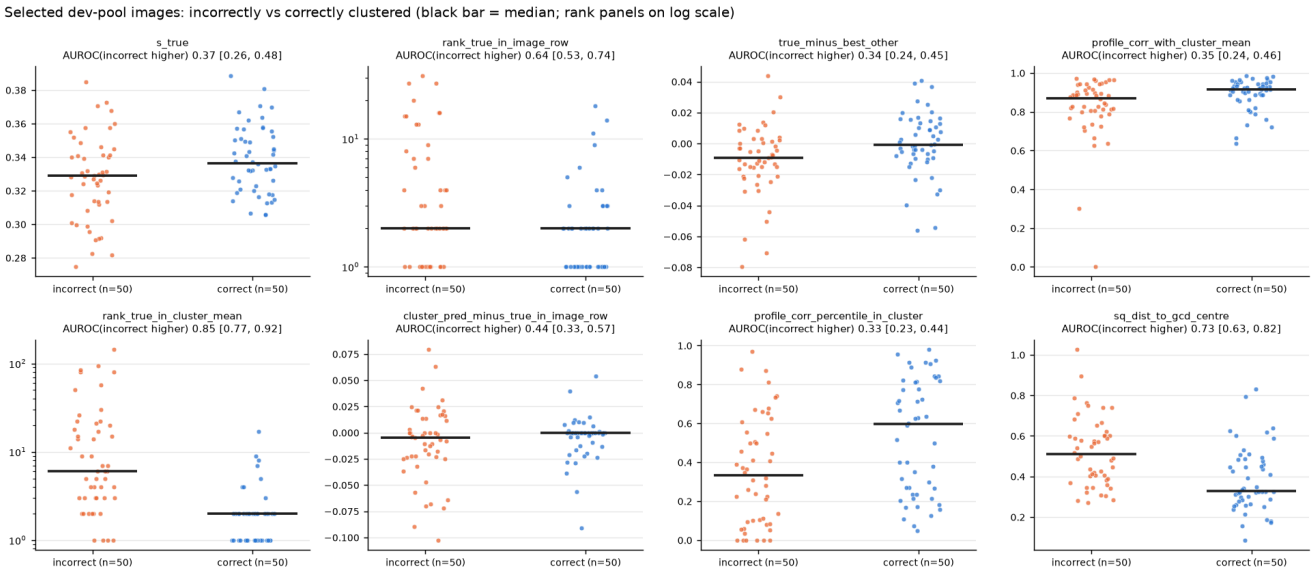
Averaged over all 200 target classes; images of the target class only. The "most similar" substitute is chosen by CLIP text similarity, which favours CLIP; it shows the failure, not its typical rate. Example: Belted Kingfisher clusters named "Ringed Kingfisher".

6. Correctly vs incorrectly clustered images (development pool)

To look at errors without touching the evaluation images, 1,433 held-out images of the known classes were assigned to the GCD clusters and a cluster-to-class map was rebuilt on them (accuracy 80.5%). 50 incorrectly and 50 correctly clustered images (seed 42) were exported with their full similarity rows and their clusters' matrices. Every check passed (exported files are byte-identical to the originals; the similarities reproduce the run's saved cluster scores within 2×10^{-7}).

Label-free measurement	Incorrect (median)	Correct (median)	AUROC, selected 100	AUROC, all 651 in these clusters
distance to the GCD cluster centre	0.510	0.330	0.73 [0.63, 0.82]	0.72 [0.67, 0.78]
typicality of the image's row within its cluster (percentile)	0.33	0.60	0.33 [0.23, 0.44]	0.33 [0.29, 0.37]
correlation of the image's row with its cluster's mean row	0.87	0.92	0.35 [0.24, 0.46]	0.36 [0.30, 0.43]
flatness of the row (normalised entropy)	0.68	0.64	0.66 [0.54, 0.76]	0.63 [0.56, 0.69]
gap between the two highest similarities	0.0044	0.0089	0.37 [0.27, 0.48]	0.41 [0.35, 0.48]
cluster: agreement among member rows	0.76	0.80	0.46 [0.34, 0.57]	0.45 [0.37, 0.54]
cluster: peakedness of the mean row	0.0058	0.0059	0.52 [0.41, 0.64]	0.46 [0.37, 0.55]

AUROC above 0.5: higher for incorrectly clustered images; below 0.5: lower. Right column: every development image in the 63 exported clusters, resampling clusters. Known classes only; exploratory (19 measurements compared, no multiplicity correction).



The 100 selected images; black bars are medians. Panels using the true label (top row and "rank_true_in_cluster_mean") describe the errors but could not detect them.

Misclustered images look like boundary cases: far from the GCD centre, less like their cluster-mates, and with flatter similarity rows. The clusters containing them do not differ in spread or peakedness. These effects are moderate (AUROC 0.63–0.73) and suggest measuring density and its gradient per image rather than per cluster in the planned mean-shift experiment.

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7. Limitations

- One training seed and one dataset. Seeds 1 and 2 and ImageNet-100 have not run; no claim about variability across training runs yet.
- CUB's fine-grained names are hard for CLIP: rows are nearly flat (top-two gaps below 0.01), so results may differ on ImageNet-100.
- The development pool contains no novel classes, so Section 6 covers known-class errors only.
- The GCD paper's estimated K for CUB (231) was not verified verbatim and is not compared.

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8. Open decisions and next runs

- **Per-size calibration.** The vocabulary size is known when naming, so calibrating thresholds per size is legitimate and would fix the softmax/MCM scale problem. Proposed as a secondary analysis before ImageNet-100.
- **Mixed precision.** fp32 training scales to roughly 175–210 GPU-hours per ImageNet-100 seed on a T4 (estimate from the CUB benchmark); mixed precision to 62–75. Proposal: rerun CUB seed 0 with mixed precision and adopt it if results match within seed variation.
- **Seeds 1 and 2 on CUB** (about 10 hours each) to measure training-seed variability.

Appendix: provenance

Run `cub/gcd/seed0_3a2d202f8444` · clustering `known_c0_fcf46ef1931c` · grounding `vitb32_831109500c95` · final checkpoint sha256 `716c3768651944c1…` · code commit `811ebfb1e37340bc7d9f3d211dfa71e689327763` · Tesla T4, torch 2.11.0+cu130. Result files: `experiments/cub/`; detailed report `report/BASELINE_REPORT.md`; diagnostics `grounding/seed0_known_c0/vitb32_831109500c95/diagnostics/clustering_examples/`. GCD paper reference: Vaze et al., CVPR 2022, arXiv:2201.02609v2, Table 3.